

ABHISHEK KAR

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SUMMARY

Mechanical Engineer (MSME, 4+ years) owning full product design from concept through NPI, validation, and production ramp for semiconductor capital equipment, consumer-electronics-class precision hardware, and instrumentation. Deep in SolidWorks, Siemens NX, and CATIA, with GD&T, statistical tolerance analysis, FEA, and hands-on prototyping and fabrication.

- **Full Product Design Ownership:** Own design end to end from concept to production across 8+ programs, cutting build time 27% and rework 31%.
- **Precision and Opto-Mechanical Design:** Design opto-mechanical, electromechanical, and enclosure hardware for RF and plasma equipment.
- **Statistical Tolerance and Root Cause:** Traced an alignment failure to the datum scheme via tolerance analysis, cutting build issues 25%.
- **Hands-On Prototyping and DFM:** Build by hand via CNC, welding, and 3D printing; design for injection molding and sheet metal at volume.

SKILLS

Design and CAD: SolidWorks, Siemens NX, CATIA (V5/V6), Creo, Autodesk Inventor (Certified), full product design, opto-mechanical and enclosure design, plastic and metal part design, GD&T (ASME Y14.5), 2D drawings, specification definition

Analysis and Simulation: FEA (structural, thermal, modal), ANSYS, CFD (ANSYS Fluent), statistical tolerance analysis, tolerance stack-up, dimensional variation analysis, first-principles modeling, DOE, MATLAB, Minitab

Manufacturing and Prototyping: DFM, DFA, error-proofing, hands-on prototyping and fabrication, CNC machining, lathe, mill, welding, 3D printing, injection molding, sheet metal, high-volume manufacturing, NPI, concept to production

Quality and Collaboration: DFMEA, PFMEA, root cause and corrective action, SPC (Cp/Cpk), reliability, design validation and characterization, supplier coordination/FAI, BOM and ECO, SAP-PLM, Teamcenter, MS Office, cross-functional collaboration

WORK EXPERIENCE

L&T Technology Services (Client: Advanced Energy)

Jun 2025 - Present

Senior Mechanical Engineer

New York, USA

- Owned full product design of opto-mechanical and electromechanical assemblies and fixtures for RF and plasma semiconductor equipment in SolidWorks and Siemens NX from concept to production across 8+ programs.
- Performed statistical tolerance and dimensional variation analysis on 10+ precision assemblies and redesigned a datum scheme through root cause, cutting alignment-related build issues 25%.
- Released detailed 2D part and assembly drawings with GD&T for production and inspection and processed 15+ ECOs in SAP-PLM, improving build accuracy 20%.
- Partnered with suppliers, manufacturing, and quality to ensure producible, reliable designs, cutting supplier lead time 23%.
- Led BOM validation across 30+ builds and supported design validation, defining test criteria and root-causing hardware failures at the assembly and system levels.

University of Minnesota

Jul 2024 - Jun 2025

Mechanical Engineer

Minnesota, USA

- Developed structural, thermal, and modal FEA in ANSYS characterizing heat transfer, stress, and vibration, validating within 7% of analytical benchmarks.
- Executed CFD in ANSYS Fluent with k-epsilon turbulence closure, deriving governing equations from first principles to support thermal and fluid design decisions.

Emerson / University of Minnesota

Sep 2023 - May 2024

Product Development Engineer

Minnesota, USA

- Built and validated 11 pre-production units by hand through machining, welding, and 3D printing, iterating CAD from test data and cutting build time 17%.
- Applied DFM and DFA for machined and molded components with manufacturing and industrial design, reducing design revisions from 7 to 3 per program.
- Conducted FEA and DFMEA to optimize structural integrity and thermal performance, improving product reliability and durability.

BGR Mining and Infra Limited

Jun 2020 - Jul 2022

Mechanical Engineer

Odisha, India

- Led root-cause failure analysis and corrective action across 20+ production equipment systems, cutting incidents 15% and improving reliability and operability.
- Engineered Lean and OEE programs using Minitab to standardize processes, cutting downtime 6% and saving roughly \$50K per year.

EDUCATION

University of Minnesota

Sep 2022 - May 2024

Master of Science, Mechanical Engineering

Minnesota, USA

KIIT University

Aug 2013 - May 2017

Bachelor of Technology, Mechanical Engineering

Odisha, India